

EC824B

PCA and Clustering

PCA and Clustering

- Two popular methods for unsupervised **dimension reduction**
- Principal components analysis (PCA):

Matrix $A \in \mathbb{R}^{N \times T}$

$$\min_{A^*} \|A - A^*\|_F^2 \text{ such that } \text{rank}(A^*) \leq R \text{ } (<< \text{min}\{N, T\})$$

“Dimension” here is rank of A , i.e. the number of orthogonal, or *unique*, vectors required to explain it

- Clustering (e.g. K -means clustering):

$a \in \mathbb{R}^{N \times p}$

$$\operatorname{argmin}_S \sum_{k=1}^K \sum_{a \in S_k} \|a - \mu_k\|^2 \equiv \operatorname{argmin}_S \sum_{k=1}^K \frac{1}{n_k} \sum_{a, a' \in S_k} \|a - a'\|^2$$

“Dimension” here is number of centroids μ_k we use to summarise our data, with $K (<< N)$

1 PCA

2 Clustering

PCA

Matrix $A \in \mathbb{R}^{N \times T}$, $\|A\|_F^2$ is the Frobenius norm:

$$\|A\|_F^2 := \sum_{i=1}^N \sum_{t=1}^T A_{it}^2$$

Rank, $\text{rank}(\cdot)$, number linearly independent columns (or rows) to exactly explain A . It is the *lowest* R we can choose such that the below is true:

$$A_{it} = \sum_{r=1}^{\text{rank}(A)} \tilde{u}_{ir} \tilde{v}_{tr} = \sum_{r=1}^{\text{rank}(A)} \sigma_r u_{ir} v_{tr}$$

where $\tilde{u}$ and $\tilde{v}$ are orthogonal but of arbitrary scale, and u and v are orthonormal, such that all scale is captured by the constants σ_r .

- Orthonormal: $\sum_i u_{ir} u_{ir'} = 1$ if $r = r'$, 0 otherwise
- Orthogonal: $\sum_i \tilde{u}_{ir} \tilde{u}_{ir'} = c_r$ if $r = r'$, 0 otherwise

PCA

A nice property of the second representation,

$$A_{it} = \sum_{r=1}^{\text{rank}(A)} \sigma_r u_{ir} v_{tr},$$

is that orthonormality gives us the following property:

$$\begin{aligned} \sum_{i=1}^N \sum_{t=1}^T A_{it}^2 &= \sum_{i=1}^N \sum_{t=1}^T \left(\sum_{r=1}^{\text{rank}(A)} \sigma_r u_{ir} v_{tr} \right)^2 \\ \text{(from orthogonality)} \quad &= \sum_{i=1}^N \sum_{t=1}^T \sum_{r=1}^{\text{rank}(A)} (\sigma_r u_{ir} v_{tr})^2 = \sum_{r=1}^{\text{rank}(A)} \sigma_r^2 \sum_{i=1}^N u_{ir}^2 \sum_{t=1}^T v_{tr}^2 \\ \text{(from normality)} \quad &= \sum_{r=1}^{\text{rank}(A)} \sigma_r^2 =: \|A\|_F^2 \end{aligned}$$

PCA - Objective function

Now we can define our objective function

$$\min_{B, \text{rank}(B) \leq R} \|A - B\|_F^2 = \min_{B, \text{rank}(B) \leq R} \sum_i \sum_t (A_{it} - B_{it})^2$$

To propose a solution, introduce the singular value decomposition (SVD):

$$A = \sum_{r=1}^{\text{rank}(A)} \sigma_r u_{ir} v_{tr} ,$$

which we saw earlier. Specifically, it is the orthonormal u and v version. By convention, the “singular values”, σ_r , are in descending order with r .

Proposition (Low-rank approximation (PCA))

$$\argmin_{B, \text{rank}(B) \leq R} \|A - B\|_F^2 = \sum_{r=1}^R \sigma_r u_r v_r' \quad (=: A^*)$$

PCA - Solution

Proposition (Low-rank approximation)

$$\operatorname{argmin}_{B, \operatorname{rank}(B) \leq R} \|A - B\|_F^2 = \sum_{r=1}^R \sigma_r u_r v_r' \quad (= A^*)$$

- $\operatorname{rank}(A^*) \leq R$: This is automatic from only R terms
- Uses the terms with the R -highest singular values, σ_r
- Remember, $\sum_{r=1}^{\operatorname{rank}(A)} \sigma_r^2 = \|A\|_F^2$, so intuitively choosing the first R terms is taking the terms that contribute the most to A 's sum of squares:

$$\|A\|_F^2 = \sum_i \sum_t A_{it}^2$$

- The minimiser leaves approximation error,

$$\|A - A^*\|_F^2 = \left\| \sum_{r=1}^{\min\{N, T\}} \sigma_r u_r v_r' - \sum_{r=1}^R \sigma_r u_r v_r' \right\|_F^2 = \left\| \sum_{r=R+1}^{\min\{N, T\}} \sigma_r u_r v_r' \right\|_F^2 = \sum_{r=R+1}^{\min\{N, T\}} \sigma_r^2$$

PCA - Sketch of proof

Claim: for any $B = XY'$ where $ncol(X) = ncol(Y) = R$,

$$\|A - A^*\|_F^2 = \sum_{r=R+1}^{\min\{N, T\}} \sigma_r^2 \leq \|A - B\|_F^2$$

- By triangle inequality, if $A = A^{(1)} + A^{(2)}$, then $\sigma_1(A) \leq \sigma_1(A^{(1)}) + \sigma_1(A^{(2)})$, where $\sigma_1(\cdot)$ is the first singular value, which turns out to be a norm (hence triangle inequality applies)
- Let $A_k^{(1)}, A_k^{(2)}$ be the SVD solution of rank k for each $A^{(1)}, A^{(2)}$. Then, $\sigma_i(A^{(1)}) = \sigma_1(A^{(1)} - A_{i-1}^{(1)})$, and $\sigma_j(A^{(2)}) = \sigma_1(A^{(2)} - A_{j-1}^{(2)})$. We have,

$$\begin{aligned} \sigma_i(A^{(1)}) + \sigma_j(A^{(2)}) &\geq \sigma_1(A^{(1)} - A_{i-1}^{(1)} + A^{(2)} - A_{j-1}^{(2)}) = \sigma_1(A - A_{i-1}^{(1)} - A_{j-1}^{(2)}) \\ &\geq \sigma_1(A - A_{i+j-2}) = \sigma_{i+j-1}(A) \end{aligned}$$

The last line comes from $rank(A_{i-1}^{(1)} + A_{j-1}^{(2)}) \leq rank(A_{i+j-2})$. Substitute in $A^{(1)} = A - B$ and $A^{(2)} = B$, noting $\sigma_{R+1}(B) = 0$. Set $i \geq 1$ and $j = R + 1$ to complete the proof.

Bet on sparsity

Back to our bet on sparsity:

- For PCA, we wanted $R \ll \min\{N, T\}$: if we allow R to get too large, we admit a high-dimensional specification. In the presence of noise, may fit to noise, e.g. observe,

$$A_{it} + \varepsilon_{it}$$

then some tail terms in the SVD may explain only ε_{it} , which we don't care about

- $A \in \mathbb{R}^{N \times T}$ has $N \times T$ free variables. The low-rank approximation (PCA) has $N \times R + T \times R$ free variables. If $N, T \gg 0$ and R not too large, huge reduction:

$$\#variables\{A\} = O(NT), \quad \#variables\{A^*\} = O(\max\{N, T\})$$

PCA - Calculation

It's all well and good knowing we want the leading terms in the SVD as a low-rank approximation to A - but how do we get these?

- Remember, we want each term to be mutually orthogonal:
 - $\sum_i u_{ir} u_{ir'} = 0, \quad r \neq r'$
 - $\sum_t v_{tr} v_{tr'} = 0, \quad r \neq r'$
- Good candidate are the eigenvectors of a symmetric matrix
- Whilst A need not be symmetric, $A'A$ and AA' are!
- Luckily for us, the SVD of a matrix always exists, so we have:

$$A = U\Sigma V' \Rightarrow A'A = V\Sigma^2 V' \quad \& \quad AA' = U\Sigma^2 U',$$

where V and U are the matrices of “singular vectors” u_r and v_r we saw earlier, Σ is a diagonal matrix with σ_r along diagonal

PCA - Calculation

$$A'A = V\Sigma^2V' \quad AA' = U\Sigma^2U',$$

These are exactly eigendecompositions:

$$A'Av_r = V\Sigma^2V'v_r = \sigma_r^2v_r =: \lambda_rv_r$$

- So, v_r are the orthonormal eigenvectors of $A'A$, and σ_r are square root of the related eigenvalues (after ordering descendingly).
- u_r can be found simply by taking $Av_r = U\Sigma V'v_r = u_r\sigma_r$

Don't worry about these last few steps - software does this very efficiently! `svd(A)`

PCA - Applications

Forecasting “factor model”:

$$Y_{it} = \lambda_i' F_t + \varepsilon_{it} = \sum_{r=1}^R \lambda_{ir} F_{tr} + \varepsilon_{it}$$

e.g. i is a specific stock, t is time measurement (month, week, day, second, etc.)

- Whilst we can never identify λ and F individually, we can identify them “up-to-rotations” with the SVD low-rank approximation from above.
- Most papers skirt this issue by simply assuming the SVD identifies, up to sign, the objects of interest, using normalisations:

$$\frac{1}{T} F' F = \mathbb{I}_R \quad \lambda' \lambda = \text{diagonal},$$

so if we use the SVD on the normalised data,

$$\frac{1}{\sqrt{NT}} Y = U \Sigma V' + \frac{\varepsilon}{\sqrt{NT}} \Rightarrow F = \sqrt{T} V \quad \& \quad \lambda = \sqrt{N} U \Sigma,$$

PCA - Applications

Interactive fixed-effects models [Bai (2009); Moon and Weidner (2015)] :

$$Y_{it} = X'_{it}\beta + \lambda'_i F_t + \varepsilon_{it} = X'_{it}\beta + \sum_{r=1}^R \lambda_{ir} F_{tr} + \varepsilon_{it},$$

where X_{it} may be correlated with λ_i and F_t , then,

$$Y_{it} - X'_{it}\beta = \sum_{r=1}^R \lambda_{ir} F_{tr} + \varepsilon_{it},$$

so we can perform the SVD low-rank approximation on the residual term.

- Note: the interactive fixed-effects $\sum_{r=1}^R \lambda_{ir} F_{tr}$ is strictly more general than additive fixed-effects $\alpha_i + \gamma_t$
- Set $F_{t1} = \mathbb{1}_{T \times 1}$ and $\lambda_{i2} = \mathbb{1}_{N \times 1}$, then $\sum_{r=1}^2 \lambda_{ir} F_{tr}$ is observationally equivalent to $\alpha_i + \gamma_t$

PCA - Applications

Matrix completion (though there is a convex relaxation that is better):

$$Y_{it}(x) = \sum_{r=1}^R \lambda_{ir}(x) F_{it}(x) + \varepsilon_{it}$$

Useful for missing data problems, which we see in economics

- Causal inference: when an observation is treated ($X_{it} = 1$) we do not observe the counterfactual $Y_{it}(0)$ for that i, t - missing data problem
- Equilibrium prices, P_{it} when $Q_{it} = 0$ and no transactions are observed – though IO economists might want more economic structure to derive these!
- Bad data/selection: bad data, where for certain time periods some individuals are not observed, or selection e.g. unbalanced panels, where there is selection into observation, like taking a survey, or, attending doctor follow-up. Then $x = 0$ means data missing

PCA - Non-convex problem

The low-rank approximation (PCA) problem is “non-convex”. Why?

- Convex: For all $\{A, B\} \in \mathcal{A}$, $aA + (1 - a)B \in \mathcal{A}$ for all $a \in [0, 1]$
- The rank constraint in the PCA objective function is *not convex*:
 1. $\exists \{A, B\}$ with $\text{rank}(\{A, B\}) \leq \{R, R\}$ such that $\text{rank}(aA + (1 - a)B) > R$
 2. E.g. when $A_{11}, B_{22} \neq 0$ and $a \in (0, 1)$

$$A = \begin{bmatrix} A_{11} & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & B_{22} \end{bmatrix}, \quad aA + (1 - a)B = \begin{bmatrix} aA_{11} & 0 \\ 0 & (1 - a)B_{22} \end{bmatrix}$$

3. Then $\text{rank}(\{A, B\}) = \{1, 1\}$ but $\text{rank}(aA + (1 - a)B) = 2$
- There may be “local” optima or saddle points away from global optimum

PCA - Convex relaxation

Constrained optimisation from earlier

$$\min_{A^*, \text{rank}(A^*) \leq R} \|A - A^*\|_F^2 = \min_{A^*, \text{rank}(A^*) \leq R} \sum_i \sum_t (A_{it} - A_{it}^*)^2.$$

Consider the constraint, $\text{rank}(A^*) \leq R$, and define the *nuclear norm*

$\|A_{it}^*\|_* = \sum_{r=1}^{\text{rank}(A^*)} \sigma_r(A^*)$, and spectral norm $\|A^*\|_2 = \sigma_1(A^*)$.

- The convex relaxation of $\text{rank}(A^*) \leq R$ is then to use $\|A_{it}^*\|_* \leq c_*$
- This comes from:

$$\|A^*\|_2 \leq \gamma \Rightarrow \text{rank}(A^*) \geq \|A_{it}^*\|_* / \gamma$$

so constraining $\|A_{it}^*\|_* \leq \gamma \cdot \text{rank}(A^*)$ equivalently constrains the $\text{rank}(A^*)$

- e.g. to constrain $\text{rank}(A^*) \leq R$, use $c_* \geq \sum_{r=1}^R \sigma_r(A^*)$, assuming we only keep the leading components from SVD of A

PCA - Convex relaxation

$$\min_{A^*, \|A_{it}^*\|_* \leq c_*} \|A - A^*\|_F^2 = \min_{A^*, \|A_{it}^*\|_* \leq c_*} \sum_i \sum_t (A_{it} - A_{it}^*)^2.$$

which can be solved with a penalised optimisation:

$$\min_{A^*} \left\{ \frac{1}{2} \|A - A^*\|_F^2 + \gamma \|A^*\|_* \right\}.$$

The solution then is to use the SVD with “soft-thresholding”:

$$A_{it}^* = \sum_{r=1}^{\text{rank}(A)} (\sigma_r(A) - \gamma)_+ u_{ir} v_{tr}$$

where $(X)_+$ is X if $X > 0$ and zero otherwise (i.e. $(X)_+ = \max\{0, X\}$)

PCA - Estimation

For our purposes treat λ_i and f_t as parameters to be estimated (“fixed-effects”). A stochastic model could be written for λ_i and f_t (“random-effects”), which we won’t do.

- Objective function (for fixed R) is a non-linear least squares:

$$\begin{aligned}\{\hat{\lambda}, \hat{F}\} &= \underset{\lambda \in \mathbb{R}^{N \times R}, F \in \mathbb{R}^{T \times R}}{\operatorname{argmin}} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \lambda'_i F_t)^2 \\ &= \underset{\lambda \in \mathbb{R}^{N \times R}, F \in \mathbb{R}^{T \times R}}{\operatorname{argmin}} \|Y - \lambda F'\|_F^2\end{aligned}$$

- Notice, as before, λ and f cannot be individually identified, because for any invertible A we have the following observational equivalence:

$$\tilde{\lambda} = \lambda A, \quad \tilde{F} = F(A^{-1})', \quad \tilde{\lambda} \tilde{F}' = \lambda F'$$

- Normalise (as above): $\frac{1}{T} F' F = \mathbb{I}_R$, and $\lambda' \lambda = \text{diagonal}$

PCA - Estimation

$$\text{FOCs:} \quad \{\lambda\} : Y\hat{F} = \hat{\lambda}\hat{F}'\hat{F}, \quad \{f\} : Y'\hat{\lambda} = \hat{F}\hat{\lambda}'\hat{\lambda}$$

Pre-multiply by Y' , respectively Y , and plug one into the other:

$$Y'Y\hat{F} = \hat{F}B, \quad YY'\hat{\lambda} = \hat{\lambda}B' \tag{1}$$

Where $B = (\hat{\lambda}'\hat{\lambda})(\hat{F}'\hat{F}) \in \mathbb{R}^{R \times R}$. Assumed implicitly: $\hat{\lambda}$ and $\hat{F}$ are full-rank (no zero columns)

- Hence, (1) shows $\hat{F}$ are R eigenvectors of $Y'Y$, respectively $\hat{\lambda}$ of YY'
- Further inspection (second order conditions) shows these are the eigenvectors corresponding to the largest eigenvalues - exactly the proposition we saw earlier

PCA - Estimation asymptotic theory

[Bai (2003)] show for $\sqrt{T}/N \rightarrow 0$ and $\sqrt{N}/T \rightarrow 0$ and $N, T \rightarrow \infty$:

$$\sqrt{N} \left(\hat{F}_t - H' F_t \right) \xrightarrow{d} N(0, V_f)$$

$$\sqrt{T} \left(\hat{\lambda}_i - H^{-1} \lambda_i \right) \xrightarrow{d} N(0, V_\lambda)$$

Where H is an $R \times R$ matrix that depends on how $\hat{F}_t$ and $\hat{\lambda}_i$ are normalised (this is not B from last slide)

- Notice the scaling for the empirical process of $\hat{F}_t$ is $\sqrt{N}$ - this is because the observations $i = 1, \dots, N$ provide the identifying variation to estimate $\hat{F}_t$, likewise for $\hat{\lambda}_i$
- Variation in the “other” dimension is the identifying variation for each of these fixed-effect parameters. This is the same intuition as in the additive fixed-effect model

1 PCA

2 Clustering

K-means clustering - objective function

Suppose we have a scalar $a_i \in \mathbb{R}^N$ that we would like to reduce the dimension of

- Clustering is an effective way to map $\mathbb{R}^N \mapsto \mathbb{R}^K$ where $K \ll N$, whilst maintaining useful information from the domain space $\mathbb{R}^N$:

$$\frac{1}{N} \sum_i (a_i - \tilde{a}_{c(i)})^2 = O_p(\xi_N)$$

where $\xi_N \rightarrow 0$, e.g. $\xi_N = 1/N$. $c(i)$ indicate i 's cluster

- There are many ways to cluster, today focus on K -means: $a \in \mathbb{R}^{N \times p}$

$$\operatorname{argmin}_S \sum_{k=1}^K \sum_{a \in S_k} \|a - \mu_k\|^2 \equiv \operatorname{argmin}_S \sum_{k=1}^K \frac{1}{n_k} \sum_{a, a' \in S_k} \|a - a'\|^2$$

Centroids μ_k used to summarise our data, with $K (\ll N)$

K-means clustering - algorithm

To find the optimal partitions of the data $\mathcal{S}$, in theory we would need to compare every possible grouping! This is infeasible: number of operations $\propto 2^N$

- Hence, algorithmic approaches are used to solve this

- **Lloyd's algorithm**

1. Assignment step: for iteration t , and cluster centroids μ_k , cluster assignment $c(i)$

$$c(i) = \operatorname{argmin}_k \|a_i - \mu_k^{(t)}\|, \quad \mathcal{S}_k^{(t)} = \{a_i : c(i) = k\}$$

2. Update step:

$$\mu_k^{(t+1)} = \frac{1}{|\mathcal{S}_k^{(t)}|} \sum_{i:c(i)=k} a_i$$

- No guarantee this converges to global minimum - we should try multiple starting values

K-means clustering

Let's assume the Lloyd's algorithm converges to a stable solution that is "close" to global

- Stable: small perturbations of solution converge back to same solution
- "close": For $\mathcal{G}$ global solution, $\mathcal{S}$ local algorithm solution -

$$|\{i : c(i) \in \mathcal{S} \neq c(i) \in \mathcal{G}\}|/N = o_p(1)$$

slight abuse of notation, ignore the arbitrary nature of cluster labelling

We want either, with $\xi_N \rightarrow 0$,

$$\frac{1}{N} \sum_i \|a_i - \tilde{a}_{c(i)}\|^2 = O_p(\xi_N),$$

or, if finding some latent true group structure ($c^*(\cdot)$) is the object of interest,

$$\frac{1}{N} \sum_{i=1}^N \mathbb{1}\{c(i) = c^*(i)\} \rightarrow 1$$

K-means clustering - a useful approximation

With $\xi_N \rightarrow 0$, want an $\mathcal{S}$ such that

$$\frac{1}{N} \sum_i \|a_i - \tilde{a}_{c(i)}\|^2 = O_p(\xi_N),$$

That is, use K -means to approximate the true values in a lower dimension

- This is called a quantisation problem
- Take the worst case scenario of a_i uniformly distributed (unit ball is w.l.o.g.)
- The best we can do, with N observations and K groups produces an error $\approx O_p(K^{-2/d})$
- To have faster than the parametric rate ($N^{-1/2}$): $d \leq 2$ and $K = O(N^\alpha)$ s.t. error $\lesssim O_p(N^{-\alpha}) = o_p(N^{-1/2})$ for $\alpha \in (1/2, 1]$

“Curse of dimensionality”! e.g. [Bonhomme, Lamadon and Manresa (2022)]

- Essentially, for statistical inference purposes, rely on one or two dimensional a_i

Clustering - application

- Standard heterogeneous slope models can force analyst to ignore panel structure and estimate slopes individual by individual. That is, for each i :

$$Y_t = X_t\beta(i) + U_t$$

is simply a regression of $t = 1, \dots, T$ datapoints for each $i \rightarrow$ highly inefficient!

Dimension reduction on heterogeneous slope coefficient:

$$Y_{it} = X_{it}\beta_{c(i)} + U_{it}$$

or, with the quantile regression, [Zhang, Wang and Zhu (2019)]

$$Q_{Y_{it}}(\tau|X_{it}) = \alpha_i(\tau) + X_{it}\beta_{c(i,\tau)}(\tau).$$

Quantile version particularly useful: can pool information within clusters for *efficient estimation*

Clustering - application

Reduce dimensionality in nonlinear analysis with fixed-effects

$$\ln f(Y_i|X_i, \theta_0) = \sum_{t=1}^T f(Y_{it}|X_{it}, \alpha_{i0}, \theta_0) =: \Lambda_i(\alpha_{i0}|\theta_0)$$

for known “link” function $\Lambda_i(\cdot)$, e.g. Gaussian $\Phi(\cdot)$

- We still need something to cluster on
- Need some aggregator over Y_{it}, X_{it} , or outside information that is informative of α_{i0}
- E.g. “moment conditions”, moment function $h_i = \frac{1}{T} \sum_{t=1}^T Y_{it}$ s.t.

$$\frac{1}{N} \sum_{i=1}^N \|h_i - \alpha_{i0}\|^2 = O_p(1/T) \quad (= o_p(1) \text{ for } T \rightarrow \infty)$$

Aside: K-nearest neighbours

KNN is a simple form of collecting observations that are close to a_i - but it is *not* clustering

- It involves taking the K closest observations to a_i . Define,
 $\mathcal{K}(i) := \{i' \neq i : \|a_i - a_{i'}\| < K + 1 \text{ largest } \|a_i - a_{i''}\| \text{ where } i'' \neq i\}$

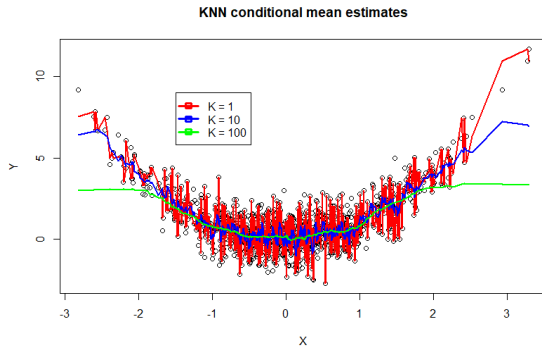
$$w_{ii',K} := \mathbb{1}\{i' \in \mathcal{K}(i)\}$$

- E.g. to estimate an average nonparametrically,

$$\hat{E}(Y_i | A_i = a_i) = \frac{1}{K} \sum_{i'=1}^N w_{ii',K} Y_{i'}$$

Aside: K-nearest neighbours

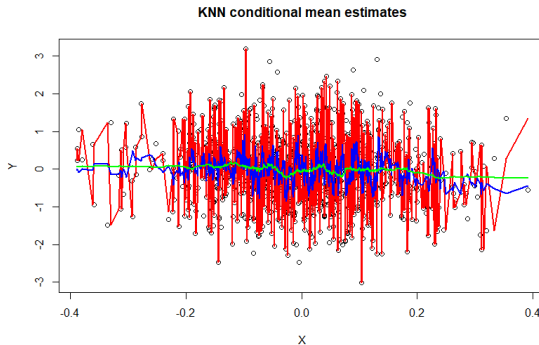
$$Y_i = X_i^2 + e_i, \quad X_i \sim N(0, 1), \quad e_i \sim N(0, 1)$$



Aside: K-nearest neighbours

Lower signal to noise ratio - still get smoother predictions for $Y_i|X_i$, but we could of course just use unconditional expectation here...

$$Y_i = X_i^2 + e_i, \quad X_i \sim N(0, 1/\sqrt{8}), \quad e_i \sim N(0, 1)$$



Aside: K-nearest neighbours

What about for our unsupervised problem of dimension reduction on a_i ?

- KNN mean for a_i would simply be

$$\tilde{a}_i = \frac{1}{K} \sum_{i'=1}^N w_{ii',K} a_{i'}$$

- But we have $\tilde{a}_i \in \mathbb{R}^N$
- The **dimension is not actually reduced**, values are just regularised, or smoothed
- This is still useful, but we cannot say we have reduced dimension

Aside: Factor model

Consider “factor model”:

$$Y_{it} = \lambda_i' F_t + \varepsilon_{it} = \sum_{r=1}^R \lambda_{ir} F_{tr} + \varepsilon_{it}$$

[Bai (2009); Moon and Weidner (2015)] with modified assumptions show, $(N, T \rightarrow \infty)$,

$$\frac{1}{N} \sum_{i=1}^N \|\hat{\lambda}_i - H_1' \lambda_i\|^2 = O_p\left(\frac{1}{\sqrt{N}}\right), \quad \frac{1}{T} \sum_{t=1}^T \|\hat{F}_t - H_2' F_t\|^2 = O_p\left(\frac{1}{\sqrt{T}}\right)$$

Use for dimension reduction, or smoothing, on the space of $\sum_{r=1}^R \lambda_{ir} F_{tr}$

1. Dimension reduction: posit $\lambda_{ir} \in \mathbb{R}^{G(N)}$, $G(N) \ll N$, and/or, $F_{tr} \in \mathbb{R}^{G(T)}$, $G(T) \ll T$
2. Smoothing: Assume neighbourhoods of λ_{ir} , respectively F_{tr} , are well-populated

Aside: Factor model - dimension reduction

Dimension reduction: posit $\lambda_{ir} \in \mathbb{R}^{G(N)}$, $G(N) \ll N$, and/or, $F_{tr} \in \mathbb{R}^{G(T)}$, $G(T) \ll T$

$$Y_{it} = \lambda_i' F_t + \varepsilon_{it} = \sum_{r=1}^R \lambda_{ir} F_{tr} + \varepsilon_{it}$$

Then we can cluster (e.g. K -means) on the initial estimates $\hat{\lambda}_i$, and $\hat{F}_t$

- Reduces dimension on λ_i , and F_t
- This is double dimension reduction:
 1. First from A_{it} to $\sum_{r=1}^R \lambda_{ir} F_{tr}$
 2. Then from $\sum_{r=1}^R \lambda_{ir} F_{tr}$ to $\sum_{r=1}^R \lambda_{c_1(i)r} F_{c_2(t)r}$
- Have gone from $N \times T$ free variables in A_{it} to $N \times R + T \times R$ down further to $G(N) \times R + G(T) \times R$
- Stricter than the factor model - but we get more sparsity!

Aside: Factor model - smoothing

Smoothing: Assume neighbourhoods of λ_{ir} , respectively F_{tr} , are well-populated

$$Y_{it} = \lambda_i' F_t + \varepsilon_{it} = \sum_{r=1}^R \lambda_{ir} F_{tr} + \varepsilon_{it}$$

Use initial estimates $\hat{\lambda}_i$, and $\hat{F}_t$ to design a weight function:

$$W_{ii'}^{(1)} = \frac{K_h(\hat{\lambda}_i, \hat{\lambda}_{i'})}{\sum_j K_h(\hat{\lambda}_i, \hat{\lambda}_j)} = \frac{k(\|\hat{\lambda}_i - \hat{\lambda}_{i'}\|/h)}{\sum_j k(\|\hat{\lambda}_i - \hat{\lambda}_j\|/h)}, \quad W_{tt'}^{(2)} = \frac{K_h(\hat{f}_t, \hat{f}_{t'})}{\sum_s K_h(\hat{f}_t, \hat{f}_s)} = \frac{k(\|\hat{f}_t - \hat{f}_{t'}\|/h)}{\sum_s k(\|\hat{f}_t - \hat{f}_s\|/h)}$$

for some kernel function with bandwidth h . Smooth out λ_i and f_t :

$$\tilde{\lambda}_i = \sum_{i'=1}^N W_{ii'}^{(1)} \lambda_{i'} \quad \tilde{f}_t = \sum_{t'=1}^T W_{tt'}^{(2)} f_{t'}$$

- No further “dimension reduction” since these are still $N \times R$ and $T \times R$ free variables

References

- Bai, J. (2003). Inferential theory for factor models of large dimensions. *Econometrica* 71(1), 135–171.
- Bai, J. (2009). Panel data models with interactive fixed effects. *Econometrica* 77(4), 1229–1279.
- Bonhomme, S., T. Lamadon, and E. Manresa (2022). Discretizing unobserved heterogeneity. *Econometrica* 90(2), 625–643.
- Moon, H. R. and M. Weidner (2015). Linear regression for panel with unknown number of factors as interactive fixed effects. *Econometrica* 83(4), 1543–1579.
- Zhang, Y., H. J. Wang, and Z. Zhu (2019). Quantile-regression-based clustering for panel data. *Journal of Econometrics* 213(1), 54–67.